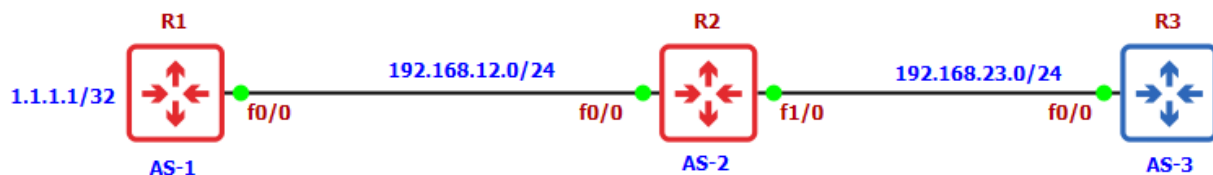


BGP Timers:

- o Basic BGP times are **Keepalive** and **Hold-down** timer intervals.
- o By default, in Border Gateway Protocol Keepalive timer is **60 seconds**.
- o By default, the BGP hold-down timer is **3 x Keepalive or 180 seconds**.
- o Once peers is UP, router starts a hold-down timer counting from 0 second up.
- o Every Keepalive message the neighbor peer resets this timer back to 0 seconds.
- o Failing to receive 3 keepalives in row make hold-down timer reach 180 seconds.
- o Neighbor is considered down and the Routes from this neighbor are flushed.
- o To verify current timers, issue the “**show ip bgp neighbor**” command in router.
- o BGP timers can be changed, both the defaults and on a per-neighbor basis.
- o In the OPEN message, BGP routers exchange the hold time they want to use.
- o Values of 1 or 2 are illegal, minimum working value for the hold time is 3 seconds.



R1 Basic Configuration

```
R1(config)#interface f0/0
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#no shutdown

R1(config)#interface loopback 1
R1(config-if)#ip add 1.1.1.1 255.255.255.255
```

R2 Basic Configuration

```
R2(config)#interface f0/0
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no shutdown

R2(config)#interface f1/0
R2(config-if)#ip add 192.168.23.2 255.255.255.0
R2(config-if)#no shutdown
```

R3 Basic Configuration

```
R3(config)#interface f0/0
R3(config-if)#ip add 192.168.23.3 255.255.255.0
R3(config-if)#no shutdown
```

Routers BGP Configuration

```
R1(config)#router bgp 1
R1(config-router)#neighbor 192.168.12.2 remote-as 2
R1(config-router)#network 1.1.1.1 mask 255.255.255.255

R2(config)#router bgp 2
R2(config-router)#neighbor 192.168.12.1 remote-as 1
R2(config-router)#neighbor 192.168.23.3 remote-as 3

R3(config)#router bgp 3
R3(config-router)#neighbor 192.168.23.2 remote-as 2
```

```
R1#show ip bgp neighbors
BGP neighbor is 192.168.12.2, remote AS 2, external link
  BGP version 4, remote router ID 192.168.23.2
  BGP state = Established, up for 00:06:56
  Last read 00:00:25, last write 00:00:40, hold time is 180, keepalive interval is 60 seconds
```

```
R2#show ip bgp neighbors
BGP neighbor is 192.168.12.1, remote AS 1, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:08:46
  Last read 00:00:39, last write 00:00:22, hold time is 180, keepalive interval is 60 seconds
```

```
R3#show ip bgp neighbors
BGP neighbor is 192.168.23.2, remote AS 2, external link
  BGP version 4, remote router ID 192.168.23.2
  BGP state = Established, up for 00:09:16
  Last read 00:00:47, last write 00:00:45, hold time is 180, keepalive interval is 60 seconds
```

R2 Configuration

```
R2(config)#router bgp 2
R2(config-router)#timers bgp 50 150
R2(config-router)#neighbor 192.168.23.3 timers 20 60
R2#clear ip bgp *
```

After changing the time for neighbor R3 new hold time is 60 and Keepalive is 20 seconds now.

```
R3#show ip bgp neighbors
BGP neighbor is 192.168.23.2, remote AS 2, external link
  BGP version 4, remote router ID 192.168.23.2
  BGP state = Established, up for 00:00:05
  Last read 00:00:04, last write 00:00:05, hold time is 60, keepalive interval is 20 seconds
```

After changing default, timers for all other new hold time is 150 & Keepalive is 50 seconds now.

```
R1#show ip bgp neighbors
BGP neighbor is 192.168.12.2, remote AS 2, external link
  BGP version 4, remote router ID 192.168.23.2
  BGP state = Established, up for 00:02:35
  Last read 00:00:15, last write 00:00:18, hold time is 150, keepalive interval is 50 seconds
```